Hyperbolic Space-Based Autoencoder for Hyperspectral Anomaly Detection

He Sun, Lizhi Wang, *Member, IEEE*, Lei Zhang, *Member, IEEE*, and Lianru Gao, *Senior Member, IEEE*

***Abstract*— Deep-learning (DL)-based methods have been shown to be effective on the hyperspectral image (HSI) anomaly detection task because of their feature extraction ability. However, current DL-based methods lack an effective means of regularizing the background information. In this article, the hyperbolic space-based autoencoder (HSAE) is proposed for the hyperspectral anomaly detection task. We assume that an effective hierarchical structural representation can better model the HSI in the spatial domain, and this enables the background information to be effectively regularized. Motivated by this idea, the HSAE embeds the HSI into hyperbolic space, which is a non-Euclidean geometry with a constant negative curvature and an exponential growth distance between points. Using a wrapped normal prior distribution, the training of the hidden representation is supervised to preserve more hierarchical features. After the training process, a hyperbolic distance-based anomaly detector (HDB) is introduced to discover anomalies in a more robust way. Experimental results on several popular HSI benchmarks fully demonstrate the superiority of our HSAE.**

***Index Terms*— Anomaly detection, autoencoder (AE), hyperbolic space, hyperspectral image (HSI).**

I. Introduction

HYPERSPECTRAL images (HSIs) can identify different material because they contain rich spectral information in a series of contiguous bands [1], [2], [3], and it has been successfully applied in a number of applications, such as anomaly detection [4]. With the aim of distinguishing infrequent anomalies that have significant differences with respect to their surrounding background, HSI anomaly detection (HAD) [5], [6] is one of the most fundamental tasks in HSI processing. Different from the target detection [7], [8], there is no prior information about anomalies, and the most significant step in the HAD task is to estimate the background information accurately [9], which helps identify anomalies and suppress interference from the background.

According to the manner of background modeling, there are three major types of HAD methods: statistical-based [4], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], representation-based [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], and deep learning (DL)-based [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52] methods. In the statistical-based methods, the background is assumed to follow a certain statistical distribution, whereas the anomalies do not. In the representation-based methods, background pixels are easily formulated by a group of primary spectral atoms or a low-rank prior, whereas the anomalies are not. However, both these methods rely heavily on manually set distribution or regularization parameters, which limits the detection accuracy.

Recently, numerous DL-based methods have been extensively employed in the HSI community owing to their strong feature representation ability [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55]. Basically, DL-based HAD approaches assume that anomalies usually cannot be reconstructed by a neural network. Hence, anomalies can be discovered using the reconstruction residual. As there is no ground truth information about anomalies in the HAD task, most DL-based methods use the HSI reconstruction as a proxy task for training a network such as an autoencoder (AE), in which the reconstruction error is adopted as the loss function (LF).

Due to the strong feature extraction ability of DL models, it may be possible to fully reconstruct anomalies after a number of training epochs, which increases the difficulty of distinguishing anomalies from the reconstruction residual. In our opinion, this is mainly because most current DL methods lack an effective method for regularizing the background information when features from both background and anomaly pixels are used in the reconstruction process. Hence, it is necessary to add some regularization to the DL model.

Generally, pixels within a local region in an HSI tend to be in the same semantic class, thus pixels in the HSI can be divided into several superpixels based on their local similarity, and the whole HSI consists of numerous superpixels to form a "pixel-superpixel-HSI" hierarchy, which is shown in Fig. 1. To this extent, anomalies tend to locate in the "pixel" level of the hierarchy as they cannot form a superpixel due to their difference with neighboring pixels, and an effective hierarchical structure representation for regularizing the DL model will be beneficial for modeling the background information. Therefore, it is very necessary to design a background regularization term to supervise the DL network and extract more spatial–spectral features.

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He Sun and Lianru Gao are with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: sunhe@aircas.ac.cn; gaolr@aircas.ac.cn).

Lizhi Wang and Lei Zhang are with the School of Computer, Beijing Institute of Technology, Beijing 100081, China (e-mail: wanglizhi@bit.edu.cn; leizhang@bit.edu.cn).

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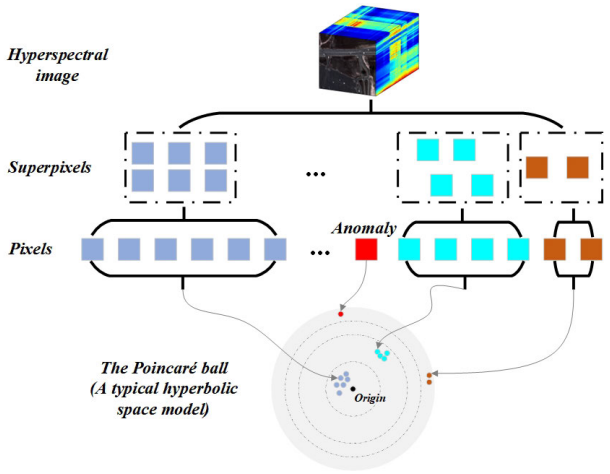


Fig. 1. Spatial hierarchy of HSI.

TABLE I
SUMMARY OF THE HSAE ARCHITECTURE

Layer	Input Channel	Output Channel	Kernel Size
Conv3×3#1	Band	4×m	3
Conv3×3#2	4×m	2×m	3
Conv3×3#3	2×m	m	3
FC#1	m	n	-
FC#2	m	n	-
FC#3	n	m	-
Conv3×3#4	m	2×m	3
Conv3×3#5	2×m	4×m	3
Conv3×3#6	4×m	Band	3

During the last several years, the hyperbolic space-based method [56] has become an effective tool for learning data representations because of its ability to represent a hierarchical structure [57], and it has been successfully applied in many applications in the computer vision [58], [59] community. As denoted in Fig. 1, the hyperbolic space-based method could be an effective way to regularize the hierarchical background of an HSI, which motivated us to develop a hyperbolic space-based AE (HSAE) for the HAD task. To be more specific, most of current methods focus on acquiring effective features on the spectral domain, and only a small local spatial region can be involved during the convolution operation. However, our proposed HSAE can extract more discriminative features by regularizing the whole spatial domain whilst preserving the feature extraction ability on the spectral domain, and these generated spectral–spatial feature can distinguish the anomaly and background pixels. As far as we know, the proposed HSAE is the first hyperbolic space-based HAD algorithm.

In the HSAE, a hyperbolic space-based DL model is used for the HAD task to better investigate the hierarchical structure in the spatial domain of an HSI. In this model, the extracted features are projected into hyperbolic space for regularization. To extract more hierarchical features, the hyperbolic wrapped normal distribution (WNB) is used to regularize the hidden representation, and a corresponding distribution loss is added to the reconstruction loss. After the convergence of the HSAE, a simple yet effective hyperbolic distance-based detector is developed to generate the anomaly map. In summary, our main contributions are as follows:

1) We propose a hyperbolic space-based method for the HAD task to explore the hierarchical structure of an HSI.
2) We adopt the hyperbolic WNB to regularize the generated hidden representation of the HSAE.
3) We develop the hyperbolic distance-based anomaly detector (HDB) by comparing the reconstruction error in a geometrical way.

The remainder of this article is organized as follows. Section II reviews the work related to the HSAE. Next, Section III describes the HSAE in detail, including the hyperbolic prior distribution, network architecture, and hyperbolic anomaly detector. In Section IV, the results of experiments on several public available datasets are presented, and the conclusion is presented in Section V.

II. Related Work

A. Statistical-Based Methods

Most statistical-based methods assume that the background consists of a large number of pixels and obeys a predefined distribution. Moreover, some variables to represent the background are estimated. Subsequently, outliers are identified as anomalies and the rest of the image is background.

For the HAD task, the Reed-Xiaoli (RX) algorithm [10] was a substantial development in the statistical-based methods. By computing the Mahalanobis distance between pixels, the RX method is capable of identifying anomalies that do not follow a multivariate Gaussian distribution. Accordingly, a number of expanded methods have been developed, including local RX [11], global RX [12], nonlinear RX [13], and kernel RX [14], among others [15], [16], [18]. However, RX-based methods are not suitable for scenes with complicated distributions, and the size of the window for calculating local statistics must be determined first, which may lower the robustness of the HAD.

To separate background and anomaly pixels, the kernelized isolation forest detection (KIFD) method [20] was proposed, which first maps the HSI into a feature space. Next, an iForest is constructed to detect the isolated pixels in the image. According to the generalized likelihood ratio test procedure, two adaptive HAD detectors [21] were proposed with an ad hoc modification, which has significantly improved the detection accuracy. However, most statistical-based methods rely heavily on the assumption of a uniform background distribution such as a Gaussian distribution, which makes them lack robustness to nonuniform background distributions.

B. Representation-Based Methods

In contrast to the statistical-based methods, representation-based approaches aim to reconstruct the background with similar behavior, and the remaining pixels are labeled as anomalies.

Both sparse representation-based [24], [25], [26] and collaborative representation-based [27], [28] methods are commonly used in the HAD task. By estimating an adaptive orthogonal background subspace, a joint sparse representation detection

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TABLE II
SUMMARY OF THE FIVE HSI DATASETS USED IN THE EXPERIMENTS

	Cat Island (CI)	Bay Chamoagne (BC)	Pavia (PA)	Gainesville (GA)	HYDICE (HY)
Number of bands	188	188	102	191	162
Wavelength range(nm)	400-2500	400-2500	430-860	400-2500	400-2500
Spatial size	150×150	100×100	150×150	100×100	80×100
Spatial resolution(m/pixel)	17.2	4.4	1.3	3.5	1.2
Anomaly pixels	19	11	68	52	17
Anomaly pixels percentage	0.08%	0.01%	0.3%	0.52%	0.2%
Sensors	AVIRIS	AVIRIS	ROSIS	AVIRIS	HYDICE
Type of Anomaly	Unknown	Unknown	Cars	Ships	Cars and Roofs

TABLE III
QUANTITATIVE RESULTS (AUC VALUES) OF THE COMPARED METHODS

Datasets	RX	EAS	KIFD	2S-GLRT	CRD	GTVLRR	PCA-TLRSR	LSDM-MoG	PTA	Auto-AD	LREN	HSAE
CI	0.9807	0.9506	0.9900	0.9646	0.9933	0.9757	0.9681	0.9722	0.9774	0.9667	0.9343	**0.9959**
BC	0.9998	0.9288	0.9983	0.9998	0.9998	0.9912	0.9837	0.9989	0.9992	0.9892	0.9040	**0.9999**
PA	0.9537	0.9135	0.8140	0.9327	0.8970	0.9818	0.9647	0.9510	0.9786	0.9857	0.8985	**0.9967**
GA	0.9512	0.8979	0.9894	0.9440	0.9118	0.9366	0.9874	0.9559	0.9560	0.9762	0.9208	**0.9978**
HY	0.9932	0.8300	0.9962	0.9387	0.9884	0.9935	0.9901	0.9914	0.9822	0.9926	0.7940	**0.9997**

TABLE IV
QUANTITATIVE RESULTS (AUC VALUES) OF THE AE-BASED METHODS

Datasets	AE	VAE	HSAE
CI	0.9783	0.9842	**0.9959**
BC	0.9899	0.9912	**0.9999**
PA	0.9232	0.9657	**0.9967**
GA	0.9559	0.9784	**0.9978**
HY	0.9957	0.9712	**0.9997**

framework [24] associates every pixel with a chosen subspace, and the residual is used to identify anomalies. Assuming that an anomaly cannot be roughly reconstructed from its neighborhood, the collaborative representation-based detection (CRD) method can discover anomaly pixels.

Using the low-rank and sparse prior characteristics of background and anomalies, low-rank and sparse matrix decomposition (LSDM)-based methods [29], [30], [31], [32], [33], [37], [38] aim to exploit the characteristics of background pixels, which helps distinguish anomalies by incorporating more spatial information. To better model the properties of a complex background, an enhanced LSDM based on a mixture of Gaussians (LSDM-MoG) [32] has been developed, in which the distribution of anomalies is better modeled. Some tensor representation-based methods [35] such as the prior-based tensor approximation (PTA) [34] have been employed because of the tensor properties of an HSI, which can better represent the 3-D structure of an HSI. Although representation-based methods have achieved certain success, they depend on manually set parameters and handcrafted priors, which may limit the performance of HAD.

C. DL-Based Methods

Because of their strong feature extraction ability, DL-based methods have attracted increasing attention in recent years, especially the AE-based methods [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51]. By applying the HSI reconstruction as a proxy task to supervise the designed network, it is assumed that the DL-based methods will focus on reconstructing the background pixels first, and anomalies can then be detected based on the reconstruction error.

Zhao et al. [41] designed a stacked denoising AE for exploiting more useful nonlinear features. To preserve the inherent structure of the original HSI, the manifold constrained AE [42] was developed to learn an effective latent representation for HAD. Although this approach improved detection accuracy, it relies on a preprocessing step to learn the hidden manifold. To force the learned latent presentation to follow a unit Gaussian distribution, a variational AE-based method [43] was developed that employs the spectral error map as the detection map. To maximize the angular similarity between the reconstructed and input data, the variational AE-based method uses the spectral angular distance as an LF. Recently, Wang et al. [46] proposed a fully convolutional AE for the HAD task called Auto-AD, which focuses on suppressing the background pixels.

Although the aforementioned AE-based methods have achieved certain success on the HAD task, the spatial hierarchical structure of HSI has been ignored, which hinders the background modeling. Lately, hyperbolic space-based methods have become popular in the deep-learning community because of their ability to effectively represent hierarchical structures [56]. Hence, we developed a hyperbolic space-based

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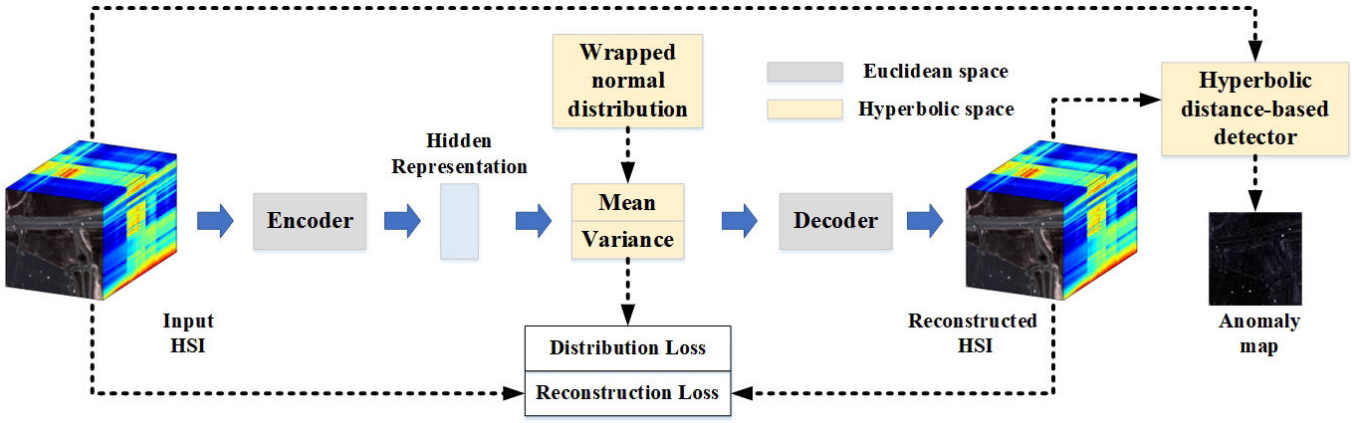


Fig. 2. HSAE framework.

TABLE V
ABLATION STUDY OF HSAE IN TERMS OF THE AUC. WNB: WRAPPED NORMAL DISTRIBUTION, FL: PROPOSED LOSS FUNCTION, AND HDB: HYPERBOLIC DISTANCE-BASED ANOMALY DETECTOR

WNB	*LF*	*HDB*	CI	BC	PA	GA	HY
			0.9649	0.9631	0.9579	0.9302	0.9192
✓			0.9693	0.9652	0.9838	0.9751	0.9571
	✓		0.9673	0.9672	0.9852	0.9752	0.9632
		✓	0.9782	0.9712	0.9733	0.9726	0.9771
✓	✓		0.9917	0.9973	0.9919	0.9938	0.9942
✓		✓	0.9909	0.9949	0.9933	0.9951	0.9957
	✓	✓	0.9917	0.9933	0.9928	0.9902	0.9966
✓	✓	✓	**0.9959**	**0.9999**	**0.9967**	**0.9978**	**0.9997**

anomaly detector, namely the HSAE, to investigate the hierarchical structure of the background.

III. HSAE

Fig. 2 presents the framework of the HSAE. A raw HSI hypercube is first taken as the input of HSAE. By regularizing the hidden features with the WNB in hyperbolic space, more hierarchical features are kept and projected back to the Euclidean space by the logarithm map. In this manner, the background pixels with a hierarchical relationship are better reconstructed than the anomalies. After the reconstruction of the HSI by the decoder, the anomaly map is generated based on a HDB, which calculates the differences between the input and reconstructed HSI in hyperbolic space. Detailed descriptions of the HSAE are given in the following.

A. Hyperbolic Prior Distribution

For AE-based HAD methods, it is very common to regularize the hidden representation for modeling the data distribution of the HSI, which causes the DL model to focus on reconstructing the background pixels. In this manner, anomalies can be detected by measuring the reconstruction error. However, the current methods are performed in Euclidean space, which cannot effectively regularize the hidden representation to investigate the spatial hierarchical structure of an HSI.

When a negative constant curvature c is set, the area of a disk in a hyperbolic space grows exponentially with respect to its radius. Similar to the growth of nodes in a tree, the hyperbolic space is thus considered to have strong potential for exploring the hierarchical structure. Therefore, the exponential growth property of the hyperbolic space is corresponding to the similar growth of leaves on a tree, and the hyperbolic space, such as the Poincaré ball model, can be assumed to be a continuous version of a tree hierarchy, which enables the hyperbolic-based DL technique can smoothly capture more hierarchical features compared to the flat Euclidean geometry. For the designed HSAE, our aim is to embed the HSI into hyperbolic space and regularize the distribution of the hidden representation, which can better suppress the background and identify anomalies.

In the HSAE, the Poincaré ball $\mathcal{P}$, a popular model in hyperbolic space, is adopted for embedding the hidden representation of the HSI. In contrast to the distance in Euclidean space $\mathcal{E}$, the hyperbolic distance d between two samples x and y in $\mathcal{P}$ is

$$d(x, y) = \cosh^{-1}\left(1 + 2\frac{\|x - y\|_2^2}{\left(1 - \|x\|_2^2\right)\left(1 - \|y\|_2^2\right) + \epsilon}\right) \quad (1)$$

where the ϵ is a small constant.

With an anchor v, the projection from the $\mathcal{E}$ to $\mathcal{P}$ relies on an exponential map, whose closed-form expression is defined as

$$\exp_x^c(v) = x \oplus_c \left(\tanh\left(\sqrt{c}\frac{\lambda_x^c\|v\|}{2}\right)\frac{v}{\sqrt{c}\|v\|}\right) \quad (2)$$

where λ_x^c equals $(2/(1 - c\|x\|^2))$. Moreover, the Möbius addition $\oplus_c$ between two points x and y on the $\mathcal{P}$ is defined as

$$x \oplus_c y = \frac{\left(1 + 2c\langle x, y\rangle + c\|y\|^2\right)x + \left(1 - c\|x\|^2\right)y}{1 + 2c\langle x, y\rangle + c^2\|x\|^2\|y\|^2} \quad (3)$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product.

For projecting back to $\mathcal{E}$, the closed-form expression of the logarithm map, i.e., the inverse of the exponential map, is

$$\log_x^c(y) = \frac{2}{\sqrt{c}\lambda_x^c}\tanh^{-1}\left(\sqrt{c}\|-x \oplus_c y\|\right)\frac{-x \oplus_c y}{\|-x \oplus_c y\|}. \quad (4)$$

To regularize the hidden distributions of the HSAE, we use the WNB, similar to [58], [59], which can map the normal distribution onto $\mathcal{E}$ via the exponential map. After sampling a point $x = \exp_\mu^c(v/\lambda_\mu^c)$ with $v \sim \mathcal{N}_\mathcal{E}(\cdot|0, \Sigma)$, the probability

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TABLE VI
FURTHER QUANTITATIVE ANALYSIS ($AUC_{t,PD}$ VALUES) OF THE COMPARED METHODS

	RX	EAS	KIFD	2S-GLRT	CRD	GTVLRR	PCA-TLRSR	LSDM-MoG	PTA	Auto-AD	LREN	HSAE
CI	0.2496	0.1367	**0.7601**	0.0950	0.4449	0.2879	0.3300	0.2004	0.2905	0.1522	0.1357	0.5006
BC	0.5314	0.5607	0.8178	0.4022	0.5709	0.6261	0.5789	0.5874	0.5798	0.3400	0.2852	**0.9677**
PA	0.1343	0.1931	0.2636	0.2525	0.1662	0.2397	0.3135	**0.4166**	0.3575	0.1063	0.0834	0.4002
GA	0.0963	0.1684	0.5729	0.2704	0.1999	0.4341	0.4605	0.2373	0.3660	0.2483	0.1974	**0.6056**
HY	0.2189	0.3051	**0.7767**	0.3103	0.2600	0.5184	0.6131	0.6587	0.6442	0.4627	0.4591	0.5995

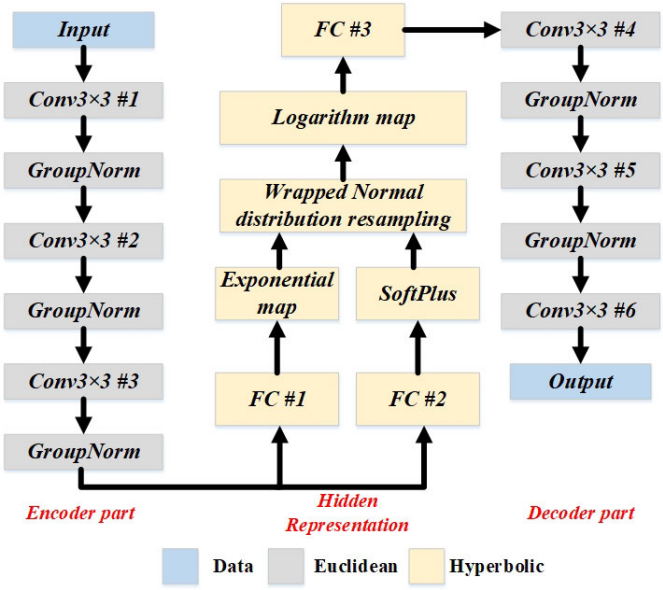


Fig. 3. Network architecture of HSAE.

density function of the WNB can be denoted using the mean μ and covariance Σ as

$$\mathcal{N}_{\mathcal{P}}(x|\mu,\Sigma)=\mathcal{N}_{\mathcal{E}}\left(\lambda_{\mu}^{c}|0,\Sigma\right)\left(\frac{d(\mu,x)}{\sinh(d(\mu,x))}\right). \tag{5}$$

Accordingly, we define a WNB $p(x)$ with mean zero as the prior, which is given by

$$p(x)=\mathcal{N}_{\mathcal{P}}(x|0,\Sigma). \tag{6}$$

By regularizing the distribution of hidden representation so that it is closer to $p(x)$, the HSAE will pay more attention to preserving the hierarchical structure of HSI, which can avoid the reconstruction of anomalies.

B. Network Architecture

Using the HSI reconstruction as a proxy task, the network architecture of the HSAE consists of an encoder branch to extract the hidden features of the HSI, as shown in Fig. 3 and Table I, a hyperbolic space-based hidden representation component for regularizing the hierarchical structure of the HSI and project regularized data back to $\mathcal{E}$, and a decoder for recovering the HSI. The details of the network architecture are introduced in the following.

1) Encoder: In the HSAE, we first normalize the input HSI $X \in R$ to the range [0, 1], and the size of input HSI is Batch $\times$ Band $\times$ Height $\times$ Width. For the DL-based HAD task, the batch is generally set to be 1 as each HSI has to be processed individually, and the Band, Height, and Width depend on the input HSI. The encoder of the HSAE contains three convolutional layers, and there is an activation function, i.e., ReLU, after each convolutional layer. Because the DL-based HAD task is usually trained on a single scene, i.e., the batch size is one, we use group normalization to replace the batch normalization layer after each activation function. To avoid setting too many parameters, we only set the output channel of the last convolutional layer as m, and the numbers of output channels of the first and second convolutional layers are set to $4 \times m$ and $2 \times m$, respectively. For simplicity, the encoder can be expressed as

$$H=\tau^{\text{en}}\left(W^{\text{en}}X+b^{\text{en}}\right) \tag{7}$$

where H is the output of the encoder; and τ^{en}, W^{en}, and b^{en} are the activation function, weight matrix, and bias vector of the encoder module, respectively.

2) Hidden Representation: After H, the output of the encoder, has been obtained, and it is processed by the hidden representation part to preserve more hierarchical features. First, we estimate the Fréchet mean u_h and distortion parameter σ_h to represent the hyperbolic variational posterior. For u_h, we implemented a fully connected layer (FC #1) to reduce the number of dimensions, and the Fréchet mean is then generated by exponential map $\exp_0^c$. A *softplus* function is used to estimate σ_h with another fully connected layer (FC #2) to reduce its dimensions. After dimensional reduction, the number of channels in both u_h and σ_h is n.

To regularize the hidden representation, a resampling scheme was implemented based on the aforementioned hyperbolic prior distribution. An anchor v_h is first sampled from the hyperbolic WNB using $v_h \sim \mathcal{N}_{\mathcal{E}}(\cdot|0,\sigma_h^2)$, and a new sampling point in the hyperbolic space r_h is obtained as follows:

$$r_h=\exp_{u_h}^{c}\left(\frac{v}{\lambda_{\mu_h}^{c}}\right). \tag{8}$$

The above reparameterization step maps the output of the encoder into hyperbolic space, which is similar to the normal distribution part in the Euclidean-based variational AE. Moreover, the above step can map the normal distribution in the Euclidean space along with hyperbolic space-based exponential map, which can generate a novel hyperbolic hidden

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TABLE VII
FURTHER QUANTITATIVE ANALYSIS ($AUC_{t,PF}$ VALUES) OF THE COMPARED METHODS

	RX	EAS	KIFD	2S-GLRT	CRD	GTVLRR	PCA-TLRSR	LSDM-MoG	PTA	Auto-AD	LREN	HSAE
CI	0.0065	0.0194	0.0845	0.0158	0.0135	0.0197	0.0243	0.0162	0.0174	**0.0014**	0.0187	0.0242
BC	0.0259	0.2346	0.1074	0.0057	0.0261	0.1334	0.0756	0.0506	0.0472	**0.0039**	0.1133	0.0700
PA	0.0233	0.0306	0.0878	0.0658	0.0636	0.0215	0.0437	0.1196	0.0303	**0.0015**	0.1425	0.0492
GA	0.0351	0.0555	0.1236	0.0323	0.0195	0.1161	0.0628	0.0462	0.0975	**0.0058**	0.1582	0.1478
HY	0.0331	0.1435	0.1350	**0.0068**	0.0507	0.0345	0.0926	0.1621	0.1368	0.0232	0.0922	0.0494

TABLE VIII
COMPUTATIONAL TIME (S) OF THE COMPARISON METHODS AND HSAE

	No.Params	CI	BC	PA	GA	HY
HSAE	0	27.36	25.52	32.89	20.47	27.31
RX	0	**0.05**	**0.07**	**0.03**	**0.03**	**0.08**
KIFD	3	26.63	24.32	28.12	28.12	25.69
2S-GLRT	2	53.52	56.79	62.12	71.32	61.28
CRD	3	45.32	49.89	37.82	41.64	35.62
GTVLRR	3	60.11	24.32	26.63	28.12	25.69
PCA-TLRSR	3	49.81	52.73	55.62	47.72	53.48
LSDM-MoG	2	38.71	35.13	29.58	37.22	33.45
PTA	3	16.72	13.41	11.06	15.79	12.11
Auto-AD	1	43.02	42.59	39.71	40.75	36.64

representation. Because r_h is sampled from the hyperbolic WNB, the newly generated hidden representation will learn a more hierarchical structure of the HSI.

After the above step has successfully projected the hidden representation into hyperbolic space and resampled a more hierarchical data point r_h, it is desirable to map r_h back to $\mathcal{E}$. In the HSAE, a logarithm map based on (4) is employed for mapping r_h to $\mathcal{E}$ and improving the dimension along with another fully connected layer (FC #3)

$$\hat{r_h} = \log_{u_h}^{c}(r_h). \tag{9}$$

3) Decoder: Using the output $\hat{r_h}$ of the hidden representation component, we employ a decoder to reconstruct the original HSI. Similar to the encoder, the decoder consists of three convolutional layers, and the input channels of the three convolutional layers are set to be m, $2 \times m$, and $4 \times m$, respectively, and the output channels are set to be $2 \times m$, and $4 \times m$, band. The group normalization has been added after the two convolutional layers. Using the decoder module, the HSI can be reconstructed as follows:

$$\tilde{X} = \tau^{\mathrm{de}}\left(W^{\mathrm{de}}\hat{r_h} + b^{\mathrm{de}}\right) \tag{10}$$

where $\tilde{X}$ is the reconstructed HSI, and τ^{de}, W^{de}, and b^{de} are the activation function, weight matrix, and bias vector of the decoder module.

4) Loss Function: Currently, most DL-based HAD methods adopt the reconstruction error as the LF L_{rec}, expressed as

$$L_{\mathrm{rec}} = \frac{1}{2S}\left\|X - \tilde{X}\right\|^2 \tag{11}$$

where S denotes all samples in HSI X.

The HSAE aims to retain more hierarchical features and avoid overfitting to the reconstructed anomalies. Because we introduce a hyperbolic prior distribution to restrict the hidden representation, a corresponding LF L_{dis}, i.e., a function comparing the distribution of H and the prior distribution $p(X)$, is defined.

Similar to the variational AE in $\mathcal{E}$, the following L_{dis} denotes the standard evidence lower bound in hyperbolic space, which is calculated by

$$\begin{aligned} L_{\mathrm{dis}} &= \int_{\mathcal{P}} \sqrt{\left\|\left(p\left(\hat{r_h}\right) - p(X)\right)\right\|^2} \\ &= \int_{\mathcal{P}} \sqrt{\left\|\mathcal{N}_{\mathcal{P}}\left(\hat{r_h}\middle|u_h, \sigma\right) - \mathcal{N}_{\mathcal{P}}(X|0, \Sigma)\right\|^2} \end{aligned} \tag{12}$$

where the $p(\hat{r_h})$ represents the prior distribution of the regularized hidden representation.

Because of the proposed distribution loss, the HSAE concentrates on learning a more hierarchical structure of the HSI, and the LF of the HSAE can be rewritten as

$$L = L_{\mathrm{rec}} + \alpha * L_{\mathrm{dis}} \tag{13}$$

where α is a coefficient for balancing L_{dis}.

C. Hyperbolic Distance-Based Anomaly Detector

Currently, most AE-based HAD methods directly use the reconstruction residual as the anomaly score. However, the above strategy cannot detect anomalies in a more structured view, which may lower the detection accuracy. Therefore, a more robust criterion for HAD is required.

Because of the regularization of the hidden representation, the HSAE will focus on recovering the background pixels and ignoring anomalies. Therefore, we propose a simple yet robust anomaly detector after the convergence of the HSAE. This detector estimates the residual E of the input and reconstructed HSI in the hyperbolic space as follows:

$$\begin{aligned} E &= d\left(X, \tilde{X}\right) \\ &= \cosh^{-1}\left(1 + 2\frac{\left\|X - \tilde{X}\right\|_2^2}{\left(1 - \|X\|_2^2\right)\left(1 - \left\|\tilde{X}\right\|_2^2\right) + \epsilon}\right). \end{aligned} \tag{14}$$

D. Implementation Details of HSAE

In this part, the implementation details of HSAE, especially the training process and the following anomaly detection procedure, will be given.

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TABLE IX
COMPUTATIONAL TIME (S) AND AUC VALUES OF HSAE AND HSAE 3-D

	CI	BC	PA	GA	HY
AUC by HSAE	0.9959	0.9999	0.9967	0.9978	0.9997
Time of HSAE	27.36	25.52	32.89	20.47	27.31
AUC by HSAE 3D	0.9912	0.9989	0.9962	0.9970	0.9995
Time of HSAE 3D	52.79	50.08	64.32	49.85	55.41

Algorithm 1 HSAE

Input: The HSI X.
Initialize: Channel numbers m and n, learning rate lr.
Begin
while training process **do**
  Perform the encoder module using Eq. (7);
  Estimate the Fréchet mean u_h and distortion parameter σ_h;
  Resample r_h from the hyperbolic WNB using Eq. (8).
  Project r_h back to $\mathcal{E}$ using (9) and obtain $\hat{r_h}$;
  Perform the decoder part using Eq. (10);
  Calculate the LF using Eq. (13);
  Optimize the HSAE during backpropagation;
end while
Generate the anomaly map using Eq. (14).
End

Similar to most DL-based HAD methods, our proposed HSAE has not split the utilized dataset into training, validation, and testing sets. As the HAD task assumes no prior information about anomalies, most of DL-based HAD methods have chosen the image reconstruction as the proxy task and utilized the reconstruction error as the supervision signal. As most HSI images are very different from each other because of its captured scenes and utilized sensors, it is not realistic to split training, validation, and testing datasets. Therefore, current DL-based HAD methods, such as Auto-AD [46], directly perform the anomaly detection process after the convergence of network training. In our work, we follow the above training strategy and detection procedure.

E. Advantages of the HSAE

We are the first to propose a hyperbolic space-based method for the HAD task, namely HSAE, and the advantages of the HSAE are summarized as follows. As the resampling strategy is in hyperbolic space, the HSAE can generate an effective embedding in hyperbolic space, where the background pixels will be the focus of the reconstruction process. This helps avoid the overfitting to the reconstructed anomalies. In addition, we propose an LF with the aid of a wrapped normal prior distribution, which encourages the HSAE to retain more hierarchical features. After the training process of HSAE, a HDB is used to generate the anomaly map, which can be directly used to explore the structural information of HSI and improve detection performance. Compared to the Euclidean space-based methods, our proposed HSAE has a unique advantage of extracting more useful spatial feature whilst preserving the ability of feature extraction on the spectral domain. The pseudocode of the full HSAE approach is presented in Algorithm 1.

IV. EXPERIMENTS

A. Datasets

In our experiments, five commonly used HSI remote sensing datasets of different landscapes were used. The key information of these five benchmarks, namely Cat Island (CI), Bay Chamoagne (BC), Pavia (PA), Gainesville (GA), and HYDICE (HY), are shown in Table II, and their pseudocolor images are presented in Fig. 4. As discussed in [4], the utilized datasets for the HAD task should have a larger spatial size, and any prior information of the anomaly should be prevented from visually inspection. However, most commonly used datasets in recent literature [4], [37], [46] are cropped from a part of the original HSI. Therefore, we have utilized the cropped datasets following other recent publications for the HAD task. To better represent these datasets, the spectral difference between the anomaly and their local background are given in Fig. 4(f)–(j), and the size of local background region is set to be 5×5.

B. Settings

To better demonstrate the performance of HSAE, 11 HAD methods were compared: RX [10], EAS [4], KIFD [20], 2S-GLRT [21], CRD [27], GTVLRR [37], PCA-TLRSR [38], LSDM-MoG [32], PTA [34], Auto-AD [46], and LREN [52]. The RX method is the most classical one in the HAD community, and both the EAS and KIFD have achieved promising performance. As a novel statistical-based approach, 2S-GLRT has become more popular recently. For the CRD, LSDM-MoG, and PTA methods, all of them are representation-based methods, which consider the complex distribution of the background and the tensor property of an HSI. Both GTVLRR and PCA-TLRSR have considered the importance of low-rank properties of background information, and Auto-AD is a recently proposed DL-based method that has achieved excellent performance. For the LREN, it is another novel DL-based method that jointly taking low-rank prior into the learning process. The experimental results of all the comparison methods were obtained using the original code from their authors, and we have made great effort to tune the parameters to obtain the best results for each of these methods.

For the hyperparameters of the HSAE, we set the learning rate of the Adam optimizer and the batch size to 0.001 and 1, respectively. We set the number of training epochs to 100 to reduce the computational burden. In addition, we set the number of hidden dimensions m and n to be 20 and 5, respectively. For the hardware, all experiments were conducted on a 16 GB RAM Intel i7-1000K CPU, and the DL-based methods were accelerated using a NVIDIA 2080 GPU with 8 GB of memory. For the software, all non-DL methods were implemented with MATLAB 2020b, and the DL-methods, including ours, were implemented using PyTorch 1.10.1.

Both the receiver operating characteristic (ROC) and the area under ROC curve (AUC) were employed to evaluate the performance of the HSAE, and the ROC curves have utilized

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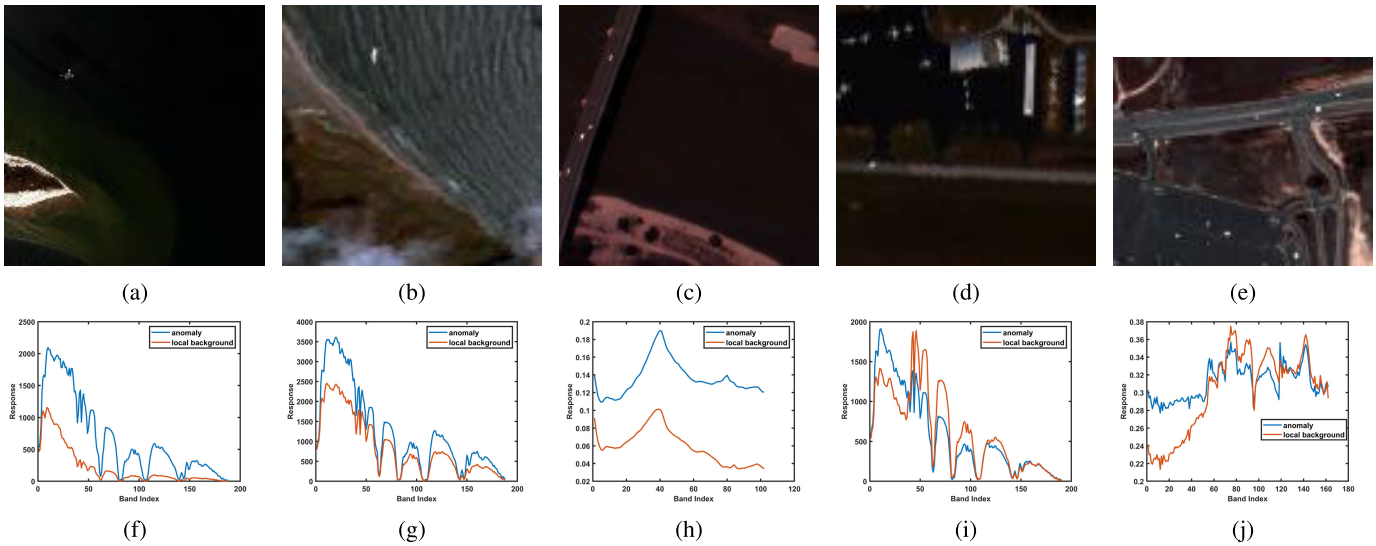


Fig. 4. Utilized datasets (a) CI, (b) BC, (c) PA, (d) GA, and (e) HI, and the spectral difference between target and local background in each dataset (f) CI, (g) BC, (h) PA, (i) GA, and (j) HI.

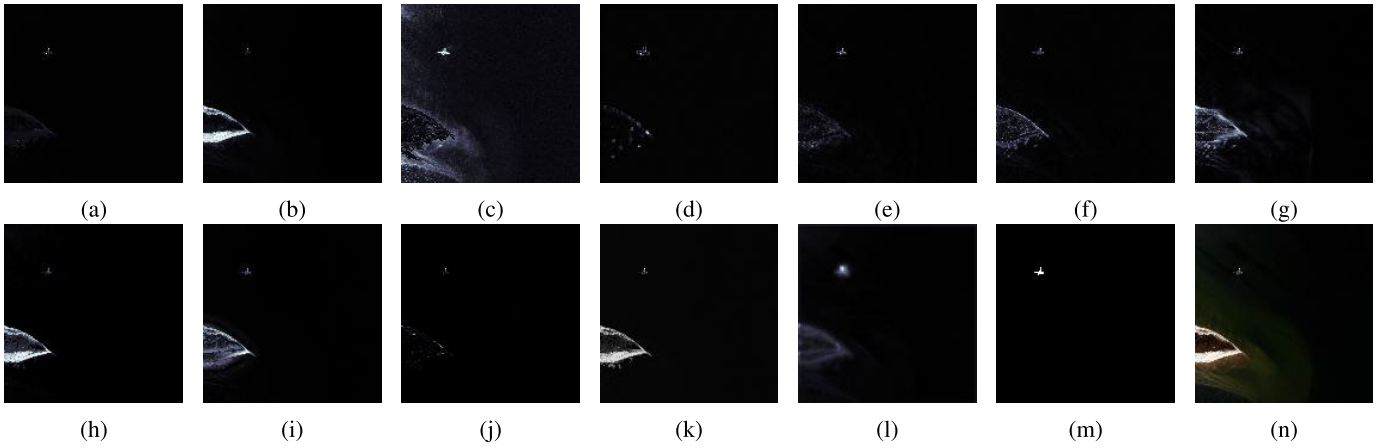


Fig. 5. Anomaly detection results for the CI dataset obtained by (a) RX, (b) EAS, (c) KIFD, (d) 2S-GLRT, (e) CRD, (f) GTVLRR, (g) PCA-TLRSR, (h) LSDM-MoG, (i) PTA, (j) Auto-AD, (k) LREN, (l) HSAE, (m) Groundtruth image, and (n) Pseudo color image of CI.

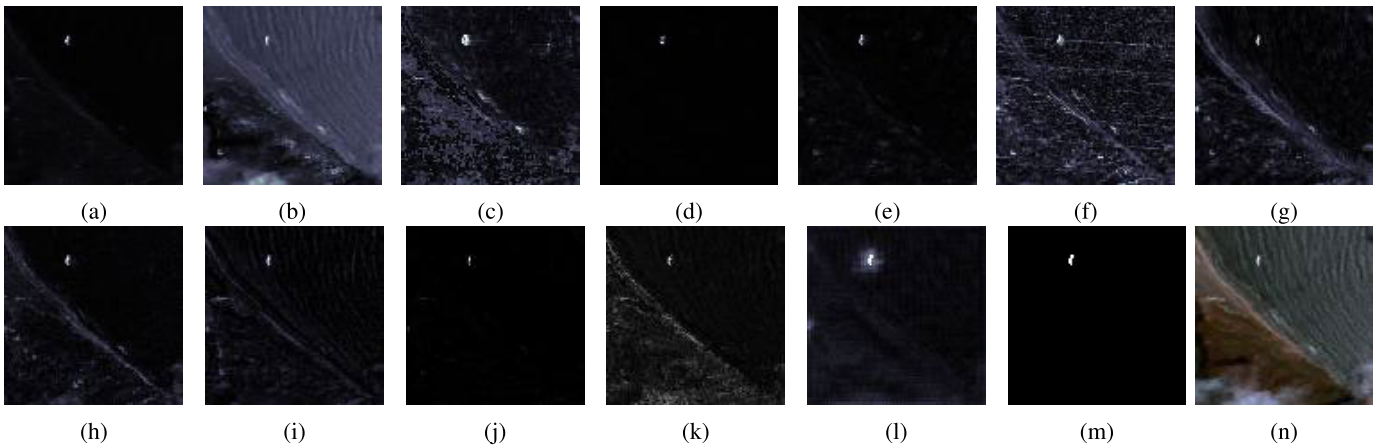


Fig. 6. Anomaly detection results for the BC dataset obtained by (a) RX, (b) EAS, (c) KIFD, (d) 2S-GLRT, (e) CRD, (f) GTVLRR, (g) PCA-TLRSR, (h) LSDM-MoG, (i) PTA, (j) Auto-AD, (k) LREN, (l) HSAE, (m) Groundtruth image, and (n) Pseudocolor image of BC.

to evaluate the detection performance with different thresholds t [60]. According to the confusion matrix, the ROC curve is generated using the true-positive rate (TPR) and false-positive rate (FPR), where the horizontal and vertical axis denote the probability of detection (PD) and false alarm rate, i.e., the probability of false detection (PF), respectively. An ROC

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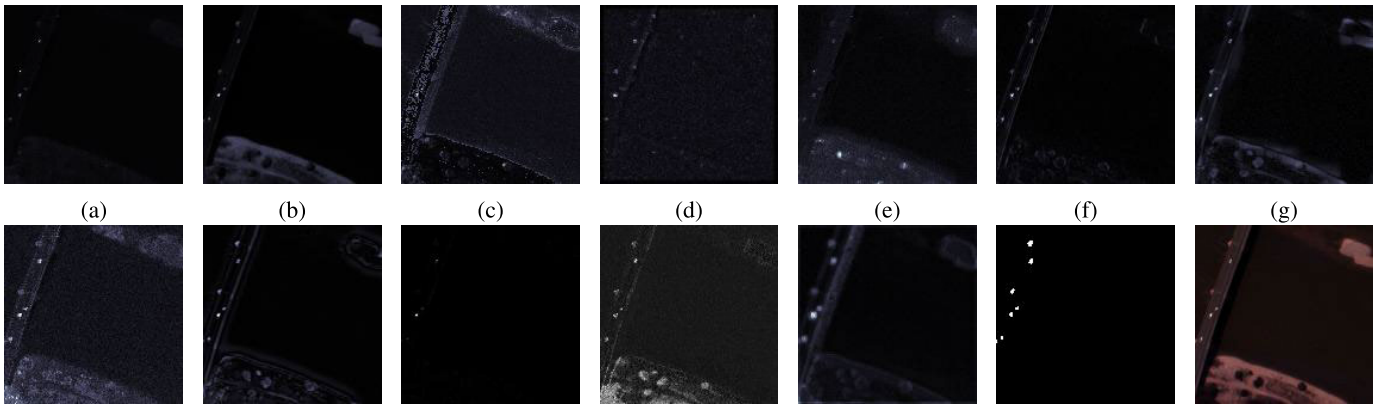


Fig. 7. Anomaly detection results for the PA dataset obtained by (a) RX, (b) EAS, (c) KIFD, (d) 2S-GLRT, (e) CRD, (f) GTVLRR, (g) PCA-TLRSR, (h) LSDM-MoG, (i) PTA, (j) Auto-AD, (k) LREN, (l) HSAE, (m) Groundtruth image, and (n) Pseudocolor image of PA.

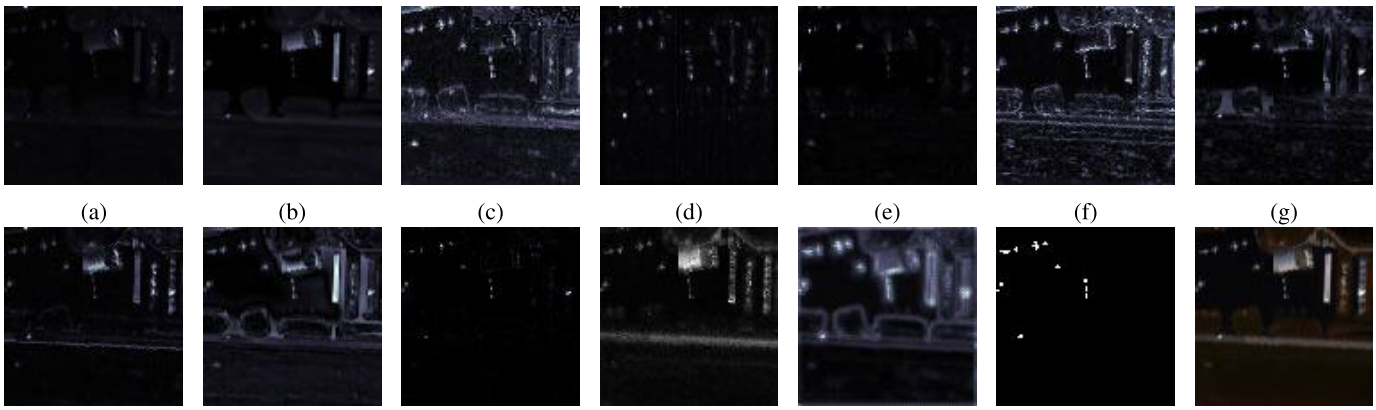


Fig. 8. Anomaly detection results for the GA dataset obtained by (a) RX, (b) EAS, (c) KIFD, (d) 2S-GLRT, (e) CRD, (f) GTVLRR, (g) PCA-TLRSR, (h) LSDM-MoG, (i) PTA, (j) Auto-AD, (k) LREN, (l) HSAE, (m) Groundtruth image, and (n) Pseudocolor image of GA.

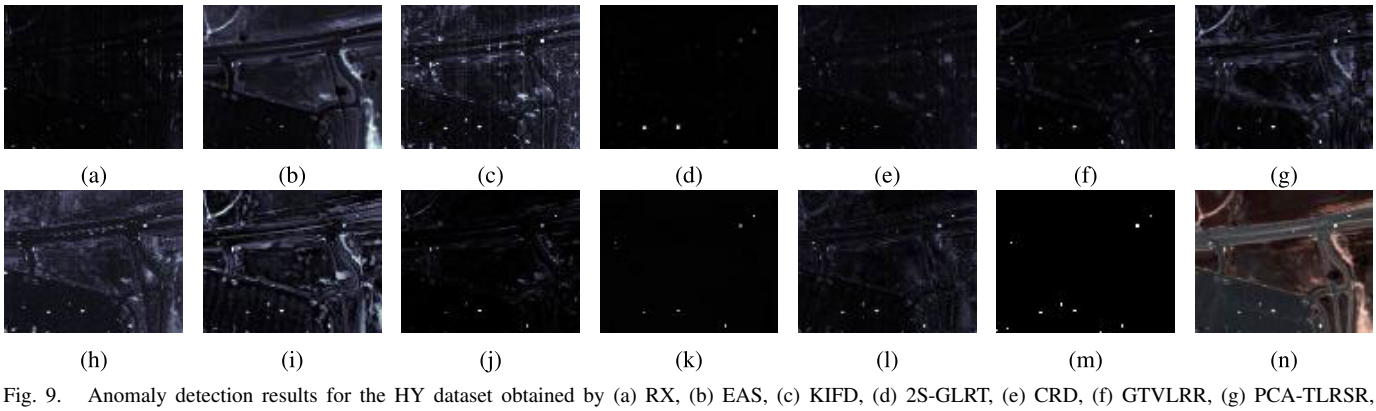


Fig. 9. Anomaly detection results for the HY dataset obtained by (a) RX, (b) EAS, (c) KIFD, (d) 2S-GLRT, (e) CRD, (f) GTVLRR, (g) PCA-TLRSR, (h) LSDM-MoG, (i) PTA, (j) Auto-AD, (k) LREN, (l) HSAE, (m) Groundtruth image, and (n) Pseudo color image of BC.

curve closer to the upper left point indicates a better detection performance. To evaluate the HAD accuracy quantitatively, a larger value of AUC indicates a better result. In the further discussion part, we have utilized the $AUC_{t,PD}$ and $AUC_{t,PF}$ to present the anomaly detection and background suppression ability based on the [4].

C. Comparison Results

The effectiveness of all methods was evaluated in three ways. The anomaly maps for visualization are given in Figs. 5–9, and the ROC curves and AUC values, which indicate detection performance, are presented in Figs. 10–14 and Table III.

The results for the CI dataset are presented in Figs. 5 and 10. Here, most methods have successfully detected the anomaly with a high score except for RX and LSDM-MoG. However, most methods have also labeled some background pixels on the left with a high anomaly intensity, and some methods even label them with higher detection probabilities, such as KIFD, GTVLRR, and PCA-TLRSR. Although Auto-AD and RX can suppress these background pixels, the anomaly pixels

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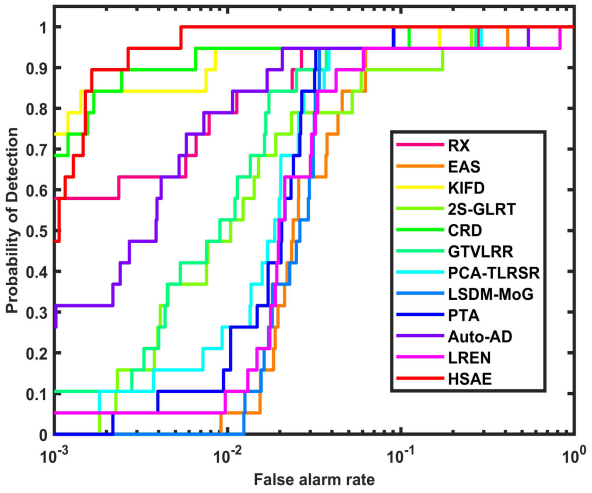


Fig. 10. Qualitative analysis of CI dataset. 2D-ROC curves of CI dataset.

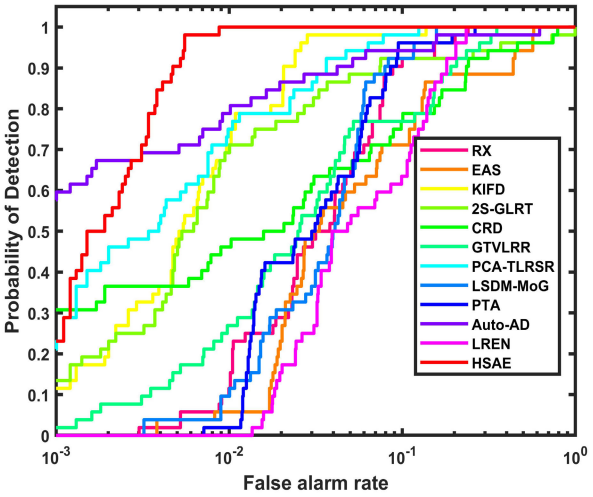


Fig. 13. Qualitative analysis of GA dataset. 2D-ROC curves of GA dataset.

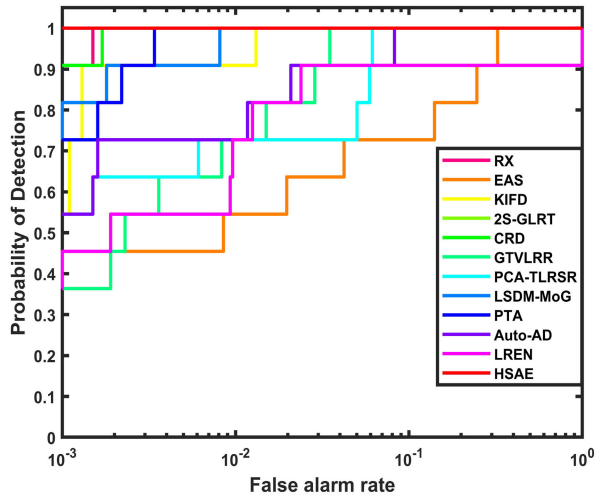


Fig. 11. Qualitative analysis of BC dataset. 2D-ROC curves of BC dataset.

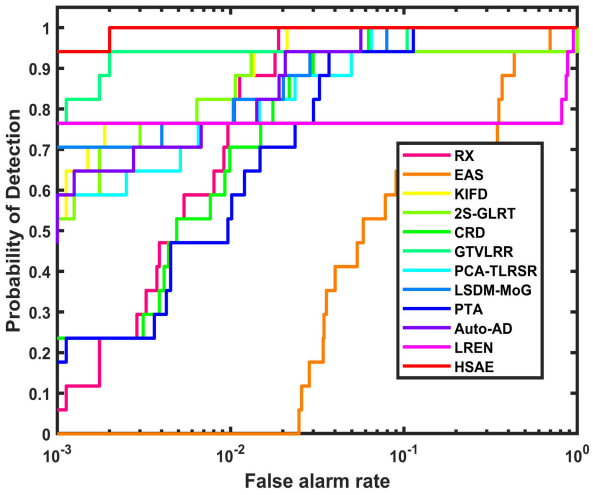


Fig. 14. Qualitative analysis of HY dataset. 2D-ROC curves of HY dataset.

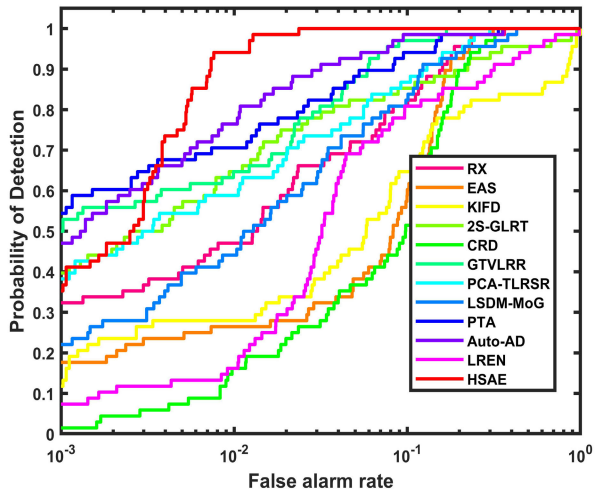


Fig. 12. Qualitative analysis of PA dataset. 2D-ROC curves of PA dataset.

have not been well identified. The results of Auto-AD reveal that it can suffer from overfitting, as some anomaly pixels are ignored because of the full reconstruction. As the Auto-AD is a kind of AE-based method, it can be argued that the HSAE is more robust against overfitting by comparing its results with those of Auto-AD. For the LREN, it fails to suppress the background information, which results a poor detection performance. The qualitative results are presented in Fig. 10, including the 2D-ROC curves and 3D-ROC curves. The HSAE has achieved a robust curve that is comparable with that of the KIFD method, while PCA-TLRSR, LSDM-MoG, and PTA have relatively poor ROC curves. From Fig. 5, it can be concluded that the HSAE can effectively separate the background and anomalies as the difference on the heat map is relatively large. According to Table III, our method has obtained the best AUC values, and all compared methods have achieved satisfactory results with AUCs higher than 0.97.

The detection performance of the BC dataset is presented in Figs. 6 and 11. In this scenario, most of the compared methods suffer from the influence of background pixels, and the HSAE has a higher detection probability than the others. KIFD has successfully identified anomalies, but its suppression of the background pixels is not as good as that of our method. Although LSDM-MoG and PTA can detect anomalies, the detection probabilities are equal to or lower than some background pixels. As for GTVLRR and PCA-TLRSR, the

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background pixels have resulted a huge influence on the detection performance. For the Auto-AD, it has effectively suppress the false alarm from background pixels thanks to its iteratively background suppression, the detection performance of anomalies has also been effected, and the detection result of LREN has suffered from poor suppression of the background same to its result of CI dataset. Fig. 11 reveals that our method has the best 2-D curve, which lies near to the top-left corner. For the heat map shown in Fig. 6, it can be seen that the detected anomalies of HSAE are rather uniform, which shows a better separability between anomalies and background. As Table III reveals, the HSAE has achieved the best AUC, which proves again the superiority of the HSAE.

Figs. 7 and 12 present the detection results of the PA dataset, in which the anomalies are some small vehicles. Here, the HSAE has performed better than the other methods. The RX method misses some anomalies, and both 2S-GLRT and CRD have failed to detect all anomalies. Although both the LSDM-MoG and PTA methods have successfully detected some targets, their results suffer from the distraction of the background pixels. For the GTVLRR and PCA-TLRSR, the false alarm from the background pixels is more serious, which large effect the detection performance. The anomaly map of the KIFD method is relatively noisy. The other DL-based algorithms, Auto-AD and LREN, has missed some anomalies, which again indicates the overfitting of DL models. Compared with the AE-based method such as Auto-AD, the HSAE has higher intensities for anomaly detection, which demonstrates its effectiveness. The ROC curves in Fig. 12 reveal that the HSAE generates a more robust curve than the compared methods. As presented in Table III, we have also achieved the highest AUC value of the methods, which proves the effectiveness of the HSAE.

Figs. 8 and 13 present the detection maps and qualitative analysis of GA dataset. Fig. 8 reveals that the HSAE can recognize several anomalies with high intensities. Apart from the 2S-GLRT and Auto-AD methods, most of the compared methods can detect the anomalies. Although KIFD performs comparably, some background pixels have higher intensities, which may lower the effectiveness of anomaly detection. Both the LSDM-MoG and PTA methods can identify some anomalies, but some background pixels heavily affect the performance, which may result a high false alarm rate. The other DL-based methods, Auto-AD and LREN, suffers from overfitting to the reconstruction of some anomalies, which affects the performance of anomaly detection. As shown in Fig. 13, the HSAE has obtained the best quantitative results of all methods, and RX and LSDM-MoG have not achieved satisfactory results. Although the Auto-AD has a better 2D-ROC curve when the false alarm rate is lower, its performance become much poorer after, where the background suppression of the Auto-AD has reduced the intensity of anomalies. Therefore, our ROC curves are still the best compared to other methods. The AUC values in Table III reveal that our method has achieved better quantitative results than KIFD and Auto-AD.

In the HY dataset, the targets are rather uniform with a simple background. As shown in Fig. 9, the HSAE, RX, Auto-AD, and AE have a better detection performance than the others, where the anomaly pixels have been well labeled. Compared with the results of the RX, PCA-TLRSR, LSDM-MoG, and PTA methods, our results show that the HSAE has suppressed most background pixels and obtained an obvious detection result. Auto-AD has reconstructed all background pixels well, but the intensities of the anomalies are lower than ours and some anomalies are not found. In the results of the remaining methods, some targets are missing or not obvious, and some anomaly maps are even very noisy. For example, KIFD has failed to reveal targets from their neighbors. Based on the results in Fig. 14, the HSAE has achieved a more reliable detection performance. The AUC values in Table III reveal that the HSAE has the best AUC and Auto-AD has obtained a similar quantitative result.

In summary, the HSAE has achieved a stable performance on five benchmarks, which indicates the robustness of the HSAE method. The KIFD method has achieved relatively good detection results on the first three datasets, but it performs poorly on the PA dataset. In contrast to the other DL-method Auto-AD, the HSAE does not reconstruct anomalies during the training process, and the robust performance of the HSAE has proved the advantage of performing the HAD task with a hyperbolic space-based DL model.

D. Further Discussions

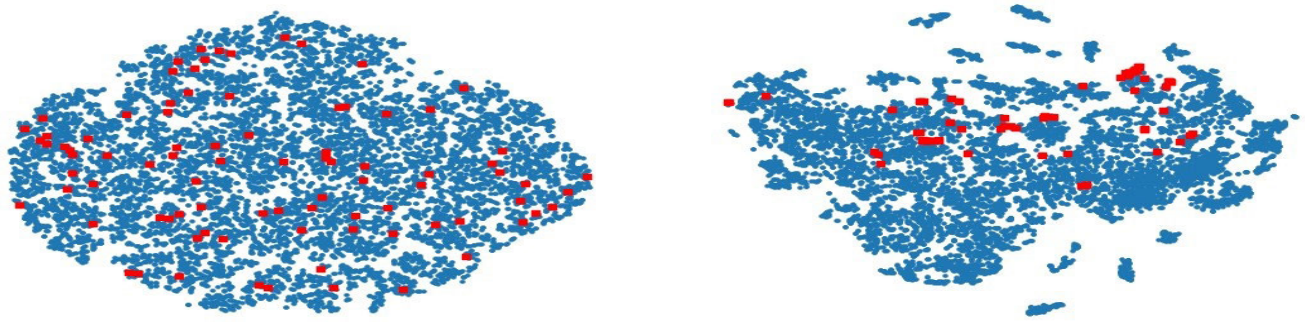
In this part, some further evaluations of the HSAE method are discussed, including the effect of hyperbolic space-based hidden representation, an ablation study to further explain the performance of the HSAE, another two parameters for further quantitative analysis, the computational time to evaluate the efficiency of HSAE, and some further discussions about the limitations of our proposed HSAE.

1) Effect of Hyperbolic Space-Based Hidden Representation: Compared to the Euclidean AE method, the HSAE method has attempted to regularize the hidden representation in the hyperbolic space. To verify the effectiveness of the HSAE, we have visualized the hidden representation with the aid of t-distributed stochastic neighbor embedding (t-SNE), which is shown in Fig. 15.

The extracted hidden representation are generated after 50 training epochs on the CI dataset, and it can be seen that the HSAE has successfully regularized the hidden representation of most anomalies. For the Euclidean AE, its hidden features of anomalies are sparsely located and very similar to the background, which has increased the difficulty of distinguishing anomalies. According to Fig. 15, the assumption that anomaly pixels are more likely to be located in the lower part of spatial hierarchy has been proved.

To further demonstrate the effectiveness of hyperbolic-based hidden representation in the HAD task, we have compared our HSAE with the Euclidean-based AE and VAE. Generally, the hidden representation of HSAE is regularized by a hyperbolic WNB, which is very similar to the Gaussian regularization in the Euclidean-based VAE. Therefore, we have conducted a comparison experiments between these three kinds of methods, where the AE and VAE are designed with same network architecture as HSAE except the hidden representation part.

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(a) (b)

Fig. 15. t-SNE visualization of hidden representation from (a) Euclidean-based and (b) Hyperbolic-based AE, the blue and red dots represent background and anomaly pixels, respectively.

For the hyperparameters, we have made great effort to achieve the best performance of AE and VAE.

As shown in Table IV, it can be seen that our HSAE has achieved the best performance compared to the AE and VAE, which has fully demonstrated the robustness of HSAE. For AE and VAE, their detection performance are not very stable, and the VAE has a better performance than AE, which proves the importance of regularizing the hidden representation part for the HAD task. Besides, the compared Auto-AD can be assumed to be an AE-based method, the superiority of our proposed HASE over the Auto-AD has fully demonstrated the effectiveness and robustness of the hyperbolic-based hidden representation.

2) Ablation Study: In this part, we evaluate the effects of the WNB, proposed LF, and HDB in Table V. For a fair comparison, we replaced these three key components with a Riemannian normal distribution, an LF with the distribution part L_{dis}, and a Euclidean distance-based anomaly detector, respectively. According to the quantitative results in the first four rows, it can be seen that the HDB component plays a more important role in improving the detection performance in most datasets, and the LF part is more significant than the WNB. After that, it is not easy to find an optimal combination of these three components until jointly applying them, which reveals the importance of jointly adopting WNB, LF, and HDB. The ablation studies have clearly showed that these three key components substantially improve the detection accuracy, and HSAE is further optimized when these three components are employed together.

According to Table V, each of the three key components in the HSAE are effective, and it can be concluded that the detection performance is further improved by applying all three components together.

3) Further Quantitative Analysis: Although the ROC curves can demonstrate the effectiveness of proposed detector, both the advantages and disadvantages of HSAE can be further discussed by analyzing two AUCs reflecting the anomaly detection and background suppression ability, i.e., the $\text{AUC}_{t,\text{PD}}$ and $\text{AUC}_{t,\text{PF}}$ based on [4]. For these two metrics, a larger $\text{AUC}_{t,\text{PD}}$ represents a better anomaly detection ability similar to the AUC, and a smaller $\text{AUC}_{t,\text{PF}}$ denotes a better background suppression ability.

In Table VI, it can be seen that our proposed HSAE has achieved a relatively good anomaly detection ability compared to other methods. Our $\text{AUC}_{t,\text{PD}}$ on both the BC and GA datasets are the best among all 11 compared methods. Although we have not achieved the best $\text{AUC}_{t,\text{PD}}$ on the CI and PA datasets, our performance are the second best, especially with a small gap on the PA dataset. For the rest methods, the KIFD has shown a better detection performance on both the CI and HY datasets, whereas its $\text{AUC}_{t,\text{PD}}$ on the PA dataset are rather poor. For the other DL-based methods, i.e. the Auto-AD, its detection performance are not as robust as ours, which corresponds to its AUC values in Table III. The background suppression ability is presented in Table VII with the $\text{AUC}_{t,\text{PF}}$. Different from the AUC values in Table III and the $\text{AUC}_{t,\text{PD}}$ values in Table VII, our HSAE has not achieved the best background suppression ability compared to others. The Auto-AD has achieved the smallest $\text{AUC}_{t,\text{PF}}$ values on four datasets, which has shown a superiority on suppressing background pixels.

Although our performance on the $\text{AUC}_{t,\text{PF}}$ is not robust enough, the HSAE has the best anomaly-background separability ability according to the quantitative analysis on Table III. The mechanism of Auto-AD is based on an iterative background suppression strategy with an adaptive-weighted loss, which mainly focus on fully reconstructing those background pixels. On the other hand, the Auto-AD has not achieved a robust anomaly detection performance due to its numerous iterations on reconstructing too much pixels, where some anomaly pixels are not effectively detected and the AUC and $\text{AUC}_{t,\text{PD}}$ values of Auto-AD are not as robust as ours. For the KIFD, its anomaly detection ability is better than the Auto-AD, but it has a larger false alarm rate than our HSAE. In this way, our HSAE can be assumed to be the most robust anomaly detection methods compared to other SOTA methods by jointly considering the effectiveness of detectors, the anomaly detection, and the background suppression ability.

4) Computational Time: To evaluate its efficiency, the HSAE is compared with other state-of-the-art methods in terms of computational time and number of key parameters in Table VIII. Because it is based on a simple detection strategy, the RX-based method is the most efficient one, and it has the advantage that no parameters need to be tuned. However, its detection performance is not as robust as that of the HSAE.

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Moreover, the KIFD, LSDM-MoG, and PTA methods have more than one manually determined key parameter. Although the computational time of the HSAE is very similar to those of the other two DL-based methods, the HSAE can be concluded to be a robust HAD method with an effective detection performance.

Despite from comparing with other state-of-the-art methods, we have investigate the trade-off between the efficiency and performance of HSAE. To be more specific, we have replaced every 2-D convolutional layers into 3-D, and the results on all five datasets are reported in Table IX. It can be seen that the performance has not been improved by introducing 3-D convolutional layers, and the computational burden is heavier than using 2-D convolutional layers. Hence, it is not necessary to introduce 3-D convolutional layer to perform the hyperspectral anomaly detection task, and what is more important is to achieve a better trade-off between the detection performance and computational efficiency.

5) Limitations and Discussions: Although our proposed HSAE has shown a robust performance on several remote sensing datasets, there are still some remained limitations for our HSAE. First, the main motivation of our method is to solve the problem of overfitting during the reconstruction process, which results some anomaly pixels have not detected. In this way, some background pixels are not effectively suppressed. In our opinion, there is a trade-off between the anomaly detection and background suppression ability, which largely effects the overall performance of HSAE. Second, the local information has not fully investigated during the reconstruction process. Our network has taken the whole HSI as an input of the network for convenience, where the local information has not fully utilized. As the anomaly usually exists sparsely in a local region, applying the whole HSI as the input may lower the performance of reconstructing background pixels, where some background pixels have not been effectively suppressed.

Through subject to the above limitations, our proposed HSAE paves a novel manner to HAD by introducing the hyperbolic space-based DL model. The strong ability of modeling hierarchical structure of the hyperbolic space-based methods has shown a better performance than the Euclidean-based ones on the HAD task, which may also provide a potential solution for other HSI processing tasks.

V. Conclusion

To investigate the spatial hierarchical structure of an HSI, a hyperbolic space-based DL model, namely HSAE, was proposed. The wrapped normal prior distribution enables the hidden representation of the HSAE to be effectively regularized, and the overfitting of reconstructing anomalies is successfully avoided. After the convergence of the proposed HSAE, a simple yet effective HDB is obtained that significantly improves the detection accuracy.

In the HSAE, we only consider the hierarchical structure of the HSI in the spatial domain, and some background pixels with significant differences between neighbors affect its detection performance. Hence, future work will focus on the following three aspects. First, a spatial–spectral DL model will be proposed to further consider the geometrical characteristic of HSIs. Second, we will attempt to collect some more suitable datasets with larger spatial size for the HAD task, where the anomaly pixels take up less portion of the whole HSI than the current utilized ones, and those datasets could be used for pretraining and fine-tuning in other DL-based HAD methods. Last, but not least, we will pay more attention on the patch-based methods to solve the problem of background suppression.

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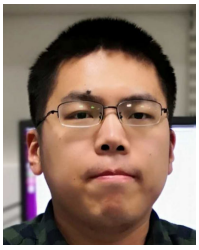
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He Sun received the B.Eng. degree in electronic and automation from Beijing Institute of Technology, Beijing, China, in 2015, the M.Sc. degree in electronic and electrical engineering from the University of Sheffield, Sheffield, U.K., in 2017, and the Ph.D. degree in electronic and electrical engineering from the University of Strathclyde, Glasgow, U.K., in 2020.

He is currently an Assistant Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include hyperspectral and optical object detection.



Lizhi Wang (Member, IEEE) received the B.S. and Ph.D. degrees from Xidian University, Xi'an, China, in 2011 and 2016, respectively.

He is currently a Professor with the School of Computer Science and Technology, Beijing Institute of Technology, Beijing, China. His research interests include computational photography and image processing.

Dr. Wang received the Best Paper Award of IEEE VCIP 2016 and the Best Paper Runner-Up Award of ACM MM 2022. He serves as an Associate Editor of IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP).



Lei Zhang (Member, IEEE) received the B.S. and Ph.D. degrees from Zhejiang University, Hangzhou, China, in 2004 and 2009, respectively.

He is currently a Professor with the School of Computer Science, Beijing Institute of Technology, Beijing, China. His research interests include image and video processing and computer graphics.



Lianru Gao (Senior Member, IEEE) received the B.S. degree in civil engineering from Tsinghua University, Beijing, China, in 2002, and the Ph.D. degree in cartography and geographic information system from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 2007.

He is currently a Professor with the Key Laboratory of Computational Optical Imaging Technology, Aerospace Information Research Institute, CAS. He also has been a Visiting Scholar at the University of Extremadura, Cáceres, Spain, in 2014, and Mississippi State University (MSU), Starkville, MI, USA, in 2016. In last ten years, he was the PI of ten scientific research projects at national and ministerial levels, including projects by the National Natural Science Foundation of China from 2018 to 2020, from 2022 to 2025, and from 2024 to 2028, and the National Key Research and Development Program of China from 2021 to 2025. He has published more than 240 peer-reviewed papers, and there are more than 150 journal articles included by Science Citation Index (SCI). He was coauthor of three academic books including *Hyperspectral Image Information Extraction*. He obtained 30 National Invention Patents in China. He was supported by the National Science Foundation for Distinguished Young Scholars of China in 2023. His research interests include hyperspectral image processing and information extraction.

Dr. Gao is a fellow of the Institution of Engineering and Technology. He received the Outstanding Science and Technology Achievement Prize of the CAS in 2016, the Second Prize of The State Scientific and Technological Progress Award in 2018, and the 2021 Outstanding Paper Award at the IEEE Workshop on Hyperspectral Image Processing: Evolution in Remote Sensing (WHISPERS). He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS) and IET *Image Processing*.

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